Daniel Felipe Vargas

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Final-year engineering student completing a double degree at École Centrale Nantes and University of the Andes, with experience in data analysis, machine learning, and software development. Seeking a six-month internship from April to September 2027.

EDUCATION

École Centrale de Nantes — Nantes, France
Diplôme d'Ingénieur (Master's level Engineering Degree) — 2025 – 2027

- Specialization in Computer Science for Information Systems

University of the Andes — Bogotá, Colombia
Bachelor's in Systems and Computer Engineering — 2021 – 2025
Minor in Computational Mathematics — GPA: 4.4 / 5.0

- National Recognition among Top-Performing Students (Colombian Ministry of Education, 2024)

EXPERIENCE

Open-Source Python Library for Affective Data Analysis — Sep 2026 – Present
École Centrale de Nantes — Nantes, France

- Developed a modular, open-source Python library enabling cognitive science researchers to quantify human emotional responses using physiological and behavioral data.
- Designed data pipelines to quantify emotional responses and variability, validating reliability against public datasets.

Climate Risk Assessment of Colombia's Hydroelectric Infrastructure — May 2026 – Sep 2026
University Jorge Tadeo Lozano — Bogotá, Colombia

- Analyzed 6 years of NASA satellite observations and climate data to assess the vulnerability of Colombia's hydroelectric infrastructure to hydroclimatic stress.
- Quantified the impact of El Niño phenomena on 23 hydroelectric plants, classifying 21% as high-risk and 30% as medium-risk.

Clinical Anomaly Detection System for Pediatric Intensive Care Units Project — Jan 2025 – Jul 2025
Central Military Hospital of Bogotá — Bogotá, Colombia

- Reduced false positive rate by 38.7% by developing real-time predictive monitoring system for pediatric vital signs, validated on 136 hours of continuous data.
- Enabled early detection of clinical deterioration by building end-to-end pipelines for data ingestion, signal processing, and anomaly detection.

Undergraduate Teaching Assistant, Software Development — Jan 2024 – Jul 2024
University of the Andes — Bogotá, Colombia

- Co-delivered a production oriented course on designing, implementing, and deploying full-stack applications using React and Spring microservices.
- Guided 10 development teams on agile methodologies, version control, test-driven development and automated deployment.

SKILLS

Programming Languages: Python, SQL, Java, C/C++, JavaScript, TypeScript, Bash

Data & Machine Learning: Signal Processing, Predictive Modeling, Anomaly Detection, Data Pipelines, Power BI

Backend & Data Systems: Spring Boot, Flask, REST APIs, Docker, PostgreSQL, MongoDB, Linux, Git, Google Cloud

LANGUAGES

Spanish – Native · **English** – C2 (Fluent) · **French** – B2 (Advanced)